Homological Algebra - Self Study

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Contents

Abstract

I am currently reading Weibel's *Introduction to Homological Algebra* to better understand homological algebra. Since I already have an understanding of some of the fundamental concepts of homological algebra, I started with chapter 7 during spring 2026 due to two of my classes working through Lie algebra cohomology. I intend to read through the book starting in chapter 1 this summer, since homology is becoming increasingly relevant in my coursework and it would be beneficial to improve my understanding of the subject. [Wei94]

Chapter 1

Lie algebra homology and cohomology

1.1 Lie algebras

Let R be a commutative ring.

Definition 1.1.1. A *Lie R-algebra*, usually called a *Lie algebra*, is an R-module $\mathfrak{g}$ together with a *Lie bracket* $[-,-] : \mathfrak{g} \otimes_R \mathfrak{g} \to \mathfrak{g}$ sometimes referred to as a multiplication operation, which is not necessarily associative, but instead satisfies the following:

- **(Antisymmetry)** $[x, x] = 0$;
- **(Jacobi identity)** $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$.

A morphism $\varphi : \mathfrak{g} \to \mathfrak{h}$ of Lie R-algebras is an R-linear map which preserves the bracket in the sense that the diagram

$$\begin{tikzcd} \mathfrak{g} \times \mathfrak{g} \arrow[r, "{[-,-]_{\mathfrak{g}}}"] \arrow[d, "\varphi \times \varphi"'] & \mathfrak{g} \arrow[d, "\varphi"] \\ \mathfrak{h} \times \mathfrak{h} \arrow[r, "{[-,-]_{\mathfrak{h}}}"'] & \mathfrak{h} \end{tikzcd}$$

commutes. One can equivalently work with maps from the tensor products, of course.

Intuitively, the bracket is a commutator, and indeed every associative R-algebra A can be viewed as a Lie R-algebra by defining $[x, y] = xy - yx$. We write $\mathrm{Lie}(A)$ for this Lie algebra, and thus we obtain a functor $\mathrm{Lie} : \mathcal{A}\mathrm{lg}_R \to \mathrm{Lie}_R$. This functor has a left adjoint given by the universal enveloping algebra $U : \mathrm{Lie}_R \to \mathcal{A}\mathrm{lg}_R$, which we will explore indepth later.

An *ideal* is a R-submodule $\mathfrak{h} \subseteq \mathfrak{g}$ such that $[\mathfrak{g}, \mathfrak{h}] \subseteq \mathfrak{h}$. An ideal is immediately a Lie subalgebra, and one can quotient by an ideal to obtain a Lie algebra. Every ideal can thus be written as a short exact sequence of Lie algebras

$$0 \longrightarrow \mathfrak{h} \longrightarrow \mathfrak{g} \longrightarrow \mathfrak{g}/\mathfrak{h} \longrightarrow 0.$$

An *abelian* Lie algebra is a Lie algebra where $[-,-] = 0$.

Example 1.1.2. Given an associative R-algebra A, we obtain the Lie algebra $\mathfrak{gl}_n(A)$ of matrices with values in A, together with important Lie subalgebras

- $\mathfrak{sl}_n(A)$: the set of traceless matrices. This becomes difficult to define when A is noncommutative.
- $\mathfrak{o}_n(A)$: the set of skew-symmetric matrices $A_{ij} = -A_{ji}$.
- $\mathfrak{t}_n(A)$: the set of upper triangular matrices.

- $\mathfrak{n}_n(A)$: the set of strictly upper triangular matrices.

Example 1.1.3. Let A a not necessarily associative R-algebra. A derivation of A into itself is a R-linear map $D : A \to A$ satisfying a Leibnitz rule

$$D(ab) = (D(a))b + a(D(b)).$$

The set $\mathrm{Der}(A)$ of derivations on A has a commutator $[D_1, D_2] = D_1 \circ D_2 - D_2 \circ D_1$, which makes $\mathrm{Der}(A)$ into a Lie algebra.

Construction 1.1.4. Given an R-module M, the *free Lie algebra* $\mathfrak{f}(M)$ is a Lie algebra containing M as a submodule which is universal with respect to turning R-linear maps into Lie algebra maps. The free Lie algebra assembles into a functor $\mathfrak{f} : \mathcal{M}\mathrm{od}_R \to \mathrm{Lie}$ with $\mathfrak{f} \dashv U$ for U the forgetful functor. The existence of such a Lie algebra follows by the adjoint functor theorem.

Construction 1.1.5. Given Lie algberas $\mathfrak{g}, \mathfrak{h}$, the product Lie algebra $\mathfrak{g} \times \mathfrak{h}$ is defined by

$$[(g, h), (g', h')] = ([g, g'], [h, h']).$$

The *lower central sequence* of a Lie algebra is the descending sequence of ideals

$$\mathfrak{g} \supseteq \mathfrak{g}^2 = [\mathfrak{g}, \mathfrak{g}] \supseteq \mathfrak{g}^3 = [\mathfrak{g}^2, \mathfrak{g}] \supseteq \cdots \supseteq \mathfrak{g}^n = [\mathfrak{g}^{n-1}, \mathfrak{g}] \supseteq \cdots .$$

A Lie algebra is *nilpotent* if $\mathfrak{g}^n = 0$ for some n. For example, $\mathfrak{n}_n(A)$ is nilpotent, and so are abelian Lie algebras.

The *derived series* of a Lie algebra $\mathfrak{g}$ is the following descending sequence of ideals

$$\mathfrak{g} \supseteq \mathfrak{g}^{(1)} = [\mathfrak{g}, \mathfrak{g}] \supseteq \mathfrak{g}^{(2)} = [\mathfrak{g}^{(1)}, \mathfrak{g}^{(1)}] \supseteq \cdots \supseteq \mathfrak{g}^{(n)} = [\mathfrak{g}^{(n-1)}, \mathfrak{g}^{(n-1)}] \supseteq \cdots .$$

We say that $\mathfrak{g}$ is *solvable* if $\mathfrak{g}^{(n)} = 0$ for some n. For example, every nilpotent Lie algebra is solvable: simply show $[\mathfrak{g}^i, \mathfrak{g}^j] \subseteq \mathfrak{g}^{i+j}$ and the statement follows inductively. The converse is not true – for example, $\mathfrak{t}_n(A)$ is solvable but not nilpotent.

1.2 $\mathfrak{g}$-Modules

Definition 1.2.1. A (left) $\mathfrak{g}$-module M is an R-module together with an R-bilinear multiplication map $\mathfrak{g} \otimes_R M \to M$ (written $x \otimes m \mapsto xm$) such that

$$[x, y]m = x(ym) - y(xm).$$

A $\mathfrak{g}$-module homomorphism is an R-linear map $\varphi : M \to N$ such that $\varphi(xm) = x\varphi(m)$. We write $\mathrm{Hom}_{\mathfrak{g}}(M, N)$ or $\mathcal{M}\mathrm{od}_{\mathfrak{g}}(M, N)$ for the set of $\mathfrak{g}$-module homomorphisms, yielding an abelian category $\mathcal{M}\mathrm{od}_{\mathfrak{g}}$.

Remark 1.2.2. Note that $\mathrm{Hom}_{\mathfrak{g}}(M, N) \subseteq \mathrm{Hom}_R(M, N)$ is an R-submodule.

Example 1.2.3. Let A be an associative R-algebra. Then any A-module M is also a $\mathrm{Lie}(A)$-module by taking the action $\mathrm{Lie}(A) \otimes_R M \to M$ to be the same as the action $A \otimes_R M \to M$.

Example 1.2.4. There is a trivial $\mathfrak{g}$-module functor from $\mathcal{M}\mathrm{od}_R \to \mathcal{M}\mathrm{od}_{\mathfrak{g}}$ given by taking an R-module M, and defining the action by $\mathfrak{g}$ to be the trivial action $0 : \mathfrak{g} \otimes_R M \to M$. It is an exact functor, and elements of its image are called *trivial* $\mathfrak{g}$*-modules*. Write this functor as $\mathbf{0}$. It has left and right adjoints.

The *invariant submodule* $M^{\mathfrak{g}}$ of a $\mathfrak{g}$-module M is the submodule

$$M^{\mathfrak{g}} = \{m \in M \mid xm = 0 \forall x \in \mathfrak{g}\}.$$

We note that $M^{\mathfrak{g}} \cong \mathrm{Hom}_{\mathfrak{g}}(R, M)$, when viewing R as a trivial R-module. This follows since R-linear maps $\varphi : R \to M$ are determined by $\varphi(1)$ since $\varphi(r) = r\varphi(1)$. To make it $\mathfrak{g}$-linear, we need $x\varphi(1) = \varphi(x1) = \varphi(0) = 0$ since R is a trivial module, hence we may only map 1 to an element of $M^{\mathfrak{g}}$. Moreover such a choice uniquely determines φ. Since it is the maximal trivial submodule of $M^{\mathfrak{g}}$, we obtain $\mathbf{0} \dashv -^{\mathfrak{g}}$. This is immediate to see: a $\mathfrak{g}$-module homomorphism $\mathbf{0}(N) \to M$ must identify $x\varphi(n) = 0$, so its image must fall into a trivial submodule, hence it factors through $M^{\mathfrak{g}}$. Moreover, any R-linear map $N \to M^{\mathfrak{g}}$ induces a map $\mathbf{0}(N) \to M$ in this way. Hence we have $\mathbf{0} \dashv -^{\mathfrak{g}}$. Additionally, since $M^{\mathfrak{g}} \cong \mathrm{Hom}_{\mathfrak{g}}(\mathbf{0}(R), M)$, invariants are a representable functor $-^{\mathfrak{g}} \cong \mathrm{Hom}_{\mathfrak{g}}(\mathbf{0}(R), -)$.

The *coinvariants* of M are the quotient $M/\mathfrak{g}M$. Note that this is also a trivial module since for any $x \in \mathfrak{g}$ and $m \in M$, we have $xm \in \mathfrak{g}M$ which is identified to 0 in the quotient. Since this is the largest quotient which is a trivial module, we can show that $-_{\mathfrak{g}} \dashv \mathbf{0}$. We will see later that $\mathcal{M}\mathrm{od}_{\mathfrak{g}}$ has enough projectives and injectives, hence we can compute the derived functors of (co)invariants.

Definition 1.2.5. Let M be a $\mathfrak{g}$-module. We write $H_\bullet(\mathfrak{g}, M)$ or $H_\bullet^{\mathrm{Lie}}(\mathfrak{g}, M)$ for the derived functors $L_\bullet(-_{\mathfrak{g}})(M)$ of $-_{\mathfrak{g}}$, and call them the *homology groups of* $\mathfrak{g}$ *with coefficients in* M. By definition, $H_0(\mathfrak{g}, M) = M_{\mathfrak{g}}$.

Similarly, write $H^\bullet(\mathfrak{g}, M)$ or $H^\bullet_{\mathrm{Lie}}(\mathfrak{g}, M)$ for the right derived functors $R^\bullet(-^{\mathfrak{g}})(M)$ of $-^{\mathfrak{g}}$, and call them the *cohomology groups of* $\mathfrak{g}$ *with coefficients in* M. By definition, $H^0(\mathfrak{g}, M) = M^{\mathfrak{g}}$.

1.3 Universal Enveloping Algebras

Definition 1.3.1. Let $\mathfrak{g}$ be a Lie R-algebra. Then the *universal enveloping algebra* $U\mathfrak{g}$ of $\mathfrak{g}$ is an associative R-algebra together with a map $i : \mathfrak{g} \hookrightarrow U\mathfrak{g}$ (which we will see later is injective, justifying the inclusion notation) which is universal in the sense that precomposition yields a natural isomorphism

$$i^* : \mathcal{A}\mathrm{lg}_R(U, -) \cong \mathrm{Lie}_R(-, \mathrm{Lie}).$$

Write $T(\mathfrak{g})$ for the tensor algebra on $\mathfrak{g}$. If $\mathfrak{g}$ is a Lie algebra, the universal enveloping algebra $U\mathfrak{g}$ of $\mathfrak{g}$ can be given as $T(\mathfrak{g})$ modulo the relation that the inclusion $i : \mathfrak{g} \hookrightarrow T(\mathfrak{g})$ be a Lie algebra homomorphism when $T(\mathfrak{g})$ has the bracket given by the commutator. Explicitly, this relation is

$$i[x, y]_{\mathfrak{g}} \sim [i(x), i(y)]_{U\mathfrak{g}} = i(x) \otimes i(y) - i(y) \otimes i(x).$$

Since $U\mathfrak{g}$ is an associative algebra, we write the tensor product $i(x) \otimes i(y)$ as xy, and refer to it as simply multiplication.

Proposition 1.3.2. *If* $\mathfrak{g}$ *is a Lie algbera, then every* $\mathfrak{g}$*-module is naturally a* $U\mathfrak{g}$*-module, and vice versa. This yields natural isomorphisms* $\mathcal{M}\mathrm{od}_{\mathfrak{g}} \cong \mathcal{M}\mathrm{od}_{U\mathfrak{g}}$.

Proof. Transposing under the tensor-hom adjunction shows that a $\mathfrak{g}$-module $\mathfrak{g} \otimes_R M \to M$ is an R-linear map $\mathfrak{g} \to \mathrm{End}_R(M)$ for some R-module M. Note that $\mathrm{End}_R(M)$ is an associative R-algebra, and the condition that $[x, y]m = x(ym) - y(xm)$ implies that this map must be a Lie algebra map $\mathfrak{g} \to \mathrm{Lie}(\mathrm{End}_R(M))$. A $U\mathfrak{g}$-module is an algebra map $U\mathfrak{g} \to \mathrm{End}_R(M)$. The statement now follows by transposing under the adjunction

$$\mathrm{Lie}_R(\mathfrak{g}, \mathrm{Lie}(\mathrm{End}_R(M))) \cong \mathcal{A}\mathrm{lg}_R(U\mathfrak{g}, \mathrm{End}_R(M)).$$

$\blacksquare$

Since the categories of associative algebras have enough projectives and injectives, proposition 1.3.2 shows that so do the categories of $\mathfrak{g}$-modules. Investigating the explicit construction shows that the action is simply given by, for $x_1, \ldots, x_n \in \mathfrak{g}$,

$$(x_1 \cdots x_n)m = x_1(\cdots(x_n m)\cdots).$$

We then impose bilinearity to cover all of $U\mathfrak{g}$.

Corollary 1.3.3. *Let M be a $\mathfrak{g}$-module. Then*

$$H_\bullet(\mathfrak{g}, M) \cong \mathrm{Tor}_\bullet^{U\mathfrak{g}}(R, M),$$

$$H^\bullet(\mathfrak{g}, M) \cong \mathrm{Ext}^\bullet_{U\mathfrak{g}}(R, M).$$

Proof. Since we have already observed that $M^{\mathfrak{g}} \cong \mathrm{Hom}_{\mathfrak{g}}(R, -) \cong \mathrm{Hom}_{U\mathfrak{g}}(R, -)$, the second statement is immediate by functoriality of derived functors.

For the first, note that the transpose of $0 : \mathfrak{g} \to R$ under $U \dashv \mathrm{Lie}$ is the unique R-algebra map $\varphi : U\mathfrak{g} \to R$ mapping $i(\mathfrak{g}) \mapsto 0$. Define the *augmentation ideal* $\mathfrak{J} = \operatorname{Ker}\varphi$ to be the kernel of this map; evidently $\mathfrak{J}$ is the two-sided ideal generated by the image $i(\mathfrak{g})$ of $\mathfrak{g}$ in $U\mathfrak{g}$.

We first show this map is epi. There may be a nice way to do this using the properties of hom-sets with epi maps and the adjunction, but I will use the explicit construction of $U\mathfrak{g}$. Since quotients distribute over direct sums, the quotient acts trivially on the factor $\mathfrak{g}^{\otimes 0} := R \subseteq T(\mathfrak{g})$, so we obtain $R \subseteq U\mathfrak{g}$. Note that if we take $\varphi|R = 1_R$ and $\varphi|\mathfrak{g} = 0$, then this satisfies the requirement that φ be an R-algebra map with $\varphi \circ i = 0$. Thus we must have $\varphi|R = 1_R$, and in particular $\operatorname{Im}\varphi = R$. Thus φ is epi.

The first isomorphism theorem now yields $R \cong U\mathfrak{g}/\mathfrak{J}$. We can also show that $U\mathfrak{g}/\mathfrak{J} \cong (U\mathfrak{g})_{\mathfrak{g}}$. Since $U\mathfrak{g}$ is a $\mathfrak{g}$-module essentially by definition, since the bracket becomes the commutator in $U\mathfrak{g}$, the $\mathfrak{g}$-coinvariants are $U\mathfrak{g}/\mathfrak{g}U\mathfrak{g}$. Since R is commutative, $\mathfrak{g}U\mathfrak{g}$ is the 2-sided ideal generated by $\mathfrak{g}^{\otimes n}$, $n \geq 1$. This is precisely the ideal generated by $\mathfrak{g}$ in $U\mathfrak{g}$, hence $\mathfrak{J} \cong \mathfrak{g}U\mathfrak{g}$. Thus $R \cong (U\mathfrak{g})_{\mathfrak{g}}$.

Now let M be any $\mathfrak{g}$-module; equivalently it is a $U\mathfrak{g}$-module. Bifunctoriality of tensor yields the isomorphisms

$$R \otimes_{U\mathfrak{g}} M \cong (U\mathfrak{g}/\mathfrak{J}) \otimes_{U\mathfrak{g}} M \cong (U\mathfrak{g})_{\mathfrak{g}} \otimes_{U\mathfrak{g}} M.$$

We claim $(U\mathfrak{g}/\mathfrak{J}) \otimes_{U\mathfrak{g}} M \cong M/\mathfrak{J}M$. First we note that since $\mathfrak{J}$ is an ideal of $U\mathfrak{g}$, there is a short exact sequence

$$0 \longrightarrow \mathfrak{J} \xhookrightarrow{j} U\mathfrak{g} \xrightarrow{\pi} U\mathfrak{g}/\mathfrak{J} \longrightarrow 0.$$

Since $- \otimes_{U\mathfrak{g}} M$ is right exact, we obtain the exact sequence

$$\mathfrak{J} \otimes_{U\mathfrak{g}} M \xrightarrow{j\otimes 1} U\mathfrak{g} \otimes_{U\mathfrak{g}} M \xrightarrow{\pi\otimes 1} U\mathfrak{g}/\mathfrak{J} \otimes_{U\mathfrak{g}} M \longrightarrow 0.$$

Note that $\operatorname{Im}(j \otimes 1) \cong \mathfrak{J} \otimes_{U\mathfrak{g}} M$, and thus exactness tells us that $\operatorname{Ker}(\pi \otimes 1) \cong \mathfrak{J} \otimes_{U\mathfrak{g}} M$. Equivalently,

$$\frac{U\mathfrak{g} \otimes_{U\mathfrak{g}} M}{\mathfrak{J} \otimes_{U\mathfrak{g}} M} \cong \operatorname{Coker}(j \otimes 1) \cong \operatorname{Coim}(\pi \otimes 1) \cong \frac{U\mathfrak{g} \otimes_{U\mathfrak{g}} M}{\operatorname{Ker}(\pi \otimes 1)} \cong \operatorname{Im}(\pi \otimes 1), .$$

where the last isomorphism is the first isomorphism theoerm, the Coker $\cong$ Coim isomorphism is exactness, and the other isomorphisms are direct constructions of the cokernel and coimage in $\mathcal{M}\mathrm{od}_{U\mathfrak{g}}$.

(The fact that $j \otimes 1$ is injective fails in general, but it's reconciled by passing to $U\mathfrak{g} \otimes_{U\mathfrak{g}} M \cong M$ in the next paragraph, and I don't wanna rewrite this).

From here, we use the fact that $U\mathfrak{g} \otimes_U \mathfrak{g}M \cong M$ is given by mapping $x \otimes m \mapsto xm$. This is an isomorphism since we take rings to be unital and ring homomorphisms to be unital, and thus $1m = m$. This guarentees a nontrivial action, and from here verification is easy. We now compute

$$\frac{U\mathfrak{g} \otimes_{U\mathfrak{g}} M}{\mathfrak{J} \otimes_{U\mathfrak{g}} M} \cong \frac{M}{\mathfrak{J}M}.$$

We now obtain that

$$\frac{M}{\mathfrak{J}M} \cong \mathrm{Im}(\pi \otimes 1) \cong \mathrm{Ker}(0 : U\mathfrak{g}/\mathfrak{J} \otimes_{U\mathfrak{g}} M \to 0) \cong U\mathfrak{g}/\mathfrak{J} \otimes_{U\mathfrak{g}} M.$$

However, since the $U\mathfrak{g}$-action on M is simply given by iterating the $\mathfrak{g}$-action on M multiple times and taking the usual R-action on M, we obtain that $M/\mathfrak{J}M \cong M/\mathfrak{g}M \cong M_{\mathfrak{g}}$. Hence we have a string of natural isomorphisms

$$R \otimes_{U\mathfrak{g}} M \cong (U\mathfrak{g}/\mathfrak{J}) \otimes_{U\mathfrak{g}} M \cong M/\mathfrak{J}M \cong M/\mathfrak{g}M \cong M_{\mathfrak{g}}.$$

Thus the derived functors of $-_{\mathfrak{g}}$ are those of $R \otimes_U \mathfrak{g}-$, which are precisely $\mathrm{Tor}_{\bullet}^{U\mathfrak{g}}(R, -)$. ■

Theorem 1.3.4 (Poincaré-Birkhoff-Witt)**.** *Let $\mathfrak{g}$ be a free R-module. Then $U\mathfrak{g}$ is also a free R-module. If $\{e_\alpha\}_{\alpha \in A}$ is an ordered basis of $\mathfrak{g}$, then the collection of all elements of the form e_I, where I is an increasing sequence, form a basis for $U\mathfrak{g}$.*

We decline to prove the PBW.

Corollary 1.3.5. *The inclusion $\mathfrak{g} \hookrightarrow U\mathfrak{g}$ is injective, so we may identify $\mathfrak{g}$ with its image.*

Corollary 1.3.6. *If $\mathfrak{h} \subseteq \mathfrak{g}$ is a Lie subalgebra, and $R = k$ is a field, then $U\mathfrak{g}$ is a free $U\mathfrak{h}$-module.*

Exercise 1.3.7. Let M, N be left $\mathfrak{g}$-modules. Show that $\mathrm{Hom}_R(M, N)$ is a $\mathfrak{g}$-module by $(xf)(m) = xf(m) - f(xm)$. Then show that $\mathrm{Hom}_{\mathfrak{g}}(M, N) \cong \mathrm{Hom}_R(M, N)^{\mathfrak{g}}$. Show that this induces isomorphisms $\mathrm{Ext}^{\bullet}_{U\mathfrak{g}}(M, N) \cong H^{\bullet}_{\mathrm{Lie}}(\mathfrak{g}, \mathrm{Hom}_R(M, N))$. In particular, the global dimension of $U\mathfrak{g}$ equals the Lie algebra cohomological dimension of $\mathfrak{g}$.

Proof. For the first, we need only show $[x, y]f = x(yf) - y(xf)$. Let $x, y \in \mathfrak{g}$ and $m \in M$. Then since M, N are $\mathfrak{g}$-modules,

$$\begin{aligned} ([x, y]f)(m) &= [x, y]f(m) - f([x, y]m) \\ &= xyf(m) - yxf(m) - f(xym - yxm) \\ &= xyf(m) - yxf(m) - f(xym) + f(yxm). \end{aligned}$$

Note that $xf = x \circ f - f \circ x$. Thus we also have

$$x(yf) = x(y \circ f - f \circ y) = x \circ y \circ f - y \circ f \circ x - x \circ f \circ y + f \circ y \circ x.$$

Hence evaluating at m yields

$$(x(yf))(m) = xyf(m) - yf(xm) - xf(ym) + f(yxm).$$

Similarly,

$$(y(xf))(m) = yxf(m) - xf(ym) - yf(xm) + f(xym).$$

Thus

$$\begin{aligned}&(x(yf))(m) - (y(xf))(m)\\ &= xyf(m) - yf(xm) - xf(ym) + f(yxm) - yxf(m) + xf(ym) + yf(xm) - f(xym)\\ &= xyf(m) + f(yxm) - yxf(m) - f(xym).\end{aligned}$$

They agree, hence $\mathrm{Hom}_R(M, N)$ is a $\mathfrak{g}$-module in the way claimed.

Now let f be a $\mathfrak{g}$-invariant of $\mathrm{Hom}_R(M, N)$. Then $0 = xf = x \circ f - f \circ x$ for all $x \in \mathfrak{g}$, and thus $x \circ f = f \circ x$. Thus f is $\mathfrak{g}$-linear. Conversely if f is $\mathfrak{g}$-linear, then $f \circ x = x \circ f$, and thus $xf = x \circ f - f \circ x = 0$, so f is a $\mathfrak{g}$-invariant of $\mathrm{Hom}_R(M, N)$. This isomorphism is clearly natural, so the claimed isomorphisms of cohomology follow. ■

1.4 H^1 and H_1

Since $\mathfrak{J}$ is the kernel of $U\mathfrak{g} \to R$, there is a short exact sequence of $\mathfrak{g}$-modules

$$0 \longrightarrow \mathfrak{J} \longrightarrow U\mathfrak{g} \longrightarrow R \longrightarrow 0.$$

Since $H_\bullet(\mathfrak{g}, -) \cong \mathrm{Tor}_\bullet^{U\mathfrak{g}}(R, -)$, we apply the *dimension shifting technique*: consider the long exact sequence

$$\cdots \longrightarrow \mathrm{Tor}_n^{U\mathfrak{g}}(U\mathfrak{g}, M) \longrightarrow \mathrm{Tor}_n^{U\mathfrak{g}}(R, M) \longrightarrow \mathrm{Tor}_{n-1}^{U\mathfrak{g}}(\mathfrak{J}, M) \longrightarrow \mathrm{Tor}_{n-1}^{U\mathfrak{g}}(U\mathfrak{g}, M) \longrightarrow \cdots.$$

Since $U\mathfrak{g}$ is free as a $U\mathfrak{g}$-module, it is flat, and hence its Tor modules vanish. Thus the short exact sequence simply reads

$$\cdots \longrightarrow 0 \longrightarrow \mathrm{Tor}_n^{U\mathfrak{g}}(R, M) \longrightarrow \mathrm{Tor}_{n-1}^{U\mathfrak{g}}(\mathfrak{J}, M) \longrightarrow 0 \longrightarrow \cdots.$$

The only way for this to be exact is if the map $\mathrm{Tor}_n^{U\mathfrak{g}}(R, M) \to \mathrm{Tor}_{n-1}^{U\mathfrak{g}}(\mathfrak{J}, M)$ is an isomorphism for $n \geq 2$. Additionally, reading this in degree $n = 1, 0$, we obtain the exact sequence

$$0 \longrightarrow H_1(\mathfrak{g}, M) \longrightarrow \mathfrak{J} \otimes_{U\mathfrak{g}} M \longrightarrow M \longrightarrow M_{\mathfrak{g}} \longrightarrow 0. \qquad (*)$$

Here we have noted that $R \otimes_{U\mathfrak{g}} M \cong M_{\mathfrak{g}}$, as computed in the proof of corollary 1.3.3.

Exercise 1.4.1. Show that the inclusion $i : \mathfrak{g} \to U\mathfrak{g}$ maps $[\mathfrak{g}, \mathfrak{g}]$ to $\mathfrak{J}^2$. Conclude that it induces a map $i : \mathfrak{g}^{ab} \to \mathfrak{J}/\mathfrak{J}^2$, where $\mathfrak{g}^{ab} = \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$.

Proof. Recall that $\mathfrak{J}$ can be described as $U\mathfrak{g} \setminus R$, which admits a description as all linear combinations in all possible n-fold products in $\mathfrak{g}$ with $n \geq 1$. Hence $\mathfrak{J}^2$ would be all linear combinations in all n-fold products in $\mathfrak{g}$ with $n \geq 2$. By definition, $i[x, y] = xy - yx$. This fits inside of $\mathfrak{J}^2$, so we obtain $i[\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{J}^2$. Thus the composite of i with the quotient $\mathfrak{J} \to \mathfrak{J}/\mathfrak{J}^2$ identifies $[\mathfrak{g}, \mathfrak{g}]$ to 0, and thus factors through the quotient $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] =: \mathfrak{g}^{ab}$. ■

Exercise 1.4.2. Show that there is a R-module map $\sigma : U\mathfrak{g} \to \mathfrak{g}^{ab}$ sending $\mathfrak{J}^2$ to zero and $i(x)$ to $\overline{x}$. *Hint:* First define a map from the tensor algebra $T(\mathfrak{g})$ to $\mathfrak{g}^{ab}$ sending $\mathfrak{g} \otimes_R \mathfrak{g}$ to zero and then pass to the quotient $U\mathfrak{g}$.

Proof. I don't know why [Wei94] denotes the class of x in the quotient by $\overline{x}$, but I hate this so I will write $i(x) \mapsto [x]$. Clearly $\sigma(x) = [x]$ would identify $[\mathfrak{g}, \mathfrak{g}]$ to 0, but showing it is well-defined, R-linear, and defining it on R is not immediate, so we start by considering the map $\widetilde{\sigma} : T(\mathfrak{g}) \to \mathfrak{g}^{ab}$ defined on $\mathfrak{g}$ by $[x]$, and defined on 2-tensors by

$$x \otimes y \mapsto [[x, y]_{\mathfrak{g}}] = 0.$$

I now understand why Weibel denotes the class by $\overline{x}$. I will still not use his notation. It is clear that $\mathfrak{g} \otimes_R \mathfrak{g} \mapsto 0$ under this map. We define it on higher tensors by

$$x_1 \otimes \cdots \otimes x_n \mapsto [[x_1, [x_2, [\cdots [x_{n-1}, x_n]_{\mathfrak{g}} \cdots]_{\mathfrak{g}}] = 0.$$

On 0-fold tensors, we define $\widetilde{\sigma}(r) = 0$. This is R-linear since the definition of the R-action on $\mathfrak{g}^{ab}$ is $r[x] = [rx]$, so $\widetilde{\sigma}(rx) = r\widetilde{\sigma}(x) = r[x] = [rx]$ holds. Since this map identifies $[x, y]_{\mathfrak{g}}$ to zero, and also identifies $x \otimes y - y \otimes x$ to zero, it must identify $[x, y]_{\mathfrak{g}} \sim x \otimes y - y \otimes x$, hence it descends to a map $\sigma : U\mathfrak{g} \to \mathfrak{g}$ defined by $x \mapsto [x]$, $R \mapsto 0$, and $\mathfrak{J}^2 \mapsto 0$. ■

Exercise 1.4.3. Deduce from exercise 1.4.1 and exercise 1.4.2 that $\mathfrak{J}/\mathfrak{J}^2 \cong \mathfrak{g}^{ab}$.

Proof. The map $i : \mathfrak{g}^{ab} \to \mathfrak{J}/\mathfrak{J}^2$ is defined by $[x] \mapsto [x]$. Consider the restriction $\sigma|\mathfrak{J} : \mathfrak{J} \to \mathfrak{g}$. Since this identifies $\mathfrak{J}^2$ to zero, it descends to a map $\overline{\sigma} : \mathfrak{J}/\mathfrak{J}^2 \to \mathfrak{g}^{ab}$ which maps $[x] \mapsto [x]$. This is an inverse to i. ■

Theorem 1.4.4. *For any Lie R-algebra $\mathfrak{g}$, we have $H_1(\mathfrak{g}, R) \cong \mathfrak{k}\mathfrak{g}^{ab}$.*

Proof. Taking $M = R$ in $(*)$ yields the exact sequence

$$0 \longrightarrow H_1(\mathfrak{g}, R) \longrightarrow \mathfrak{J} \otimes_{U\mathfrak{g}} R \longrightarrow R \longrightarrow R_{\mathfrak{g}} \longrightarrow 0.$$

Recall from the proof of corollary 1.3.3 that $R \cong U\mathfrak{g}/\mathfrak{J}$. We establish an isomorphism $\mathfrak{J} \otimes_{U\mathfrak{g}} (U\mathfrak{g}/\mathfrak{J}) \cong \mathfrak{J}/\mathfrak{J}^2$ as a special case of lemma 1.4.6, hence

$$\mathfrak{J} \otimes_{U\mathfrak{g}} R \cong \mathfrak{J} \otimes_{U\mathfrak{g}} (U\mathfrak{g}/\mathfrak{J}) \cong \mathfrak{J}/\mathfrak{J}^2 \cong \mathfrak{g}^{ab},$$

where the final isomorphism is exercise 1.4.3. We will now show the map $R \to R_{\mathfrak{g}}$ is an isomorphism. It is automatically surjective. Since $R_{\mathfrak{g}} := R/\mathfrak{g}R$, and since R has the trivial $\mathfrak{g}$-module structure,

$$\frac{R}{\mathfrak{g}R} \cong \frac{R}{0} \cong R.$$

Thus the quotient is indeed an isomorphism, meaning that

$$0 \cong \operatorname{Ker}(R \to R_{\mathfrak{g}}) \cong \operatorname{Im}(\mathfrak{J} \otimes_{U\mathfrak{g}} R \to R).$$

Hence this map is the 0 morphism, so we obtain an exact sequence

$$0 \longrightarrow H_1(\mathfrak{g}, R) \longrightarrow \mathfrak{J} \otimes_{U\mathfrak{g}} R \longrightarrow 0.$$

This verifies

$$H_1 \cong \mathfrak{J} \otimes_{U\mathfrak{g}} R \cong \mathfrak{g}^{ab}.$$

■

Corollary 1.4.5. *If M is any trivial $\mathfrak{g}$-module, then $H_1(\mathfrak{g}, M) \cong \mathfrak{g}^{ab} \otimes_R M$.*

Proof. The proof relies on $M \cong M_{\mathfrak{g}}$, so that we have an isomorphism $H_1(\mathfrak{g}, M) \cong \mathfrak{J} \otimes_{U\mathfrak{g}} M$ as in theorem 1.4.4. Now we use the isomorphism $\mathfrak{J} \otimes_{U\mathfrak{g}} R \cong \mathfrak{g}^{ab}$ to obtain

$$\mathfrak{J} \otimes_{U\mathfrak{g}} M \cong \mathfrak{J} \otimes_{U\mathfrak{g}} (R \otimes_R M) \cong (\mathfrak{J} \otimes_{U\mathfrak{g}} R) \otimes_R M \cong \mathfrak{g}^{ab} \otimes_R M.$$

■

Lemma 1.4.6. *Let A be a ring, $J \subseteq A$ an ideal, and M an A-module. Then*

$$(A/I) \otimes_A M \cong M/IM.$$

In particular, since $\mathfrak{J} \subseteq U\mathfrak{g}$ is an ideal of the R-algebra $U\mathfrak{g}$ (and in particular, an ideal as a ring), and since $\mathfrak{J}$ is thus an $U\mathfrak{g}$-module, we obtain $U\mathfrak{g}/\mathfrak{J} \otimes_{U\mathfrak{g}} \mathfrak{J} \cong \mathfrak{J}/\mathfrak{J}^2$.

We decline to prove this standard fact of algebra, but it is required for the previous two statements so we include it since we did not remember it.

Exercise 1.4.7. Let $\mathfrak{g}$ be a free R-module on basis $\{e_i\}_{1 \le i \le n}$, viewed as an abelian Lie R-algebra. Show that $H_p(\mathfrak{g}, R) \cong \Lambda^p \mathfrak{g}$.

Proof. We first find a free resolution of R over $R[e_1, \dots, e_n]$. View R as an $R[e_1, \dots, e_n]$-module with trivial action by the e_i's. Apparently the sequence is $U\mathfrak{g} \otimes_R \Lambda^p \mathfrak{g} \cong \Lambda^p$ with differential

$$d(1 \otimes x_1 \wedge \cdots \wedge x_p) = \sum_{i=1}^{p} (-1)^{i+1} x_i \otimes x_1 \wedge \cdots \wedge \widehat{x_i} \wedge \cdots \wedge x_p.$$

I will take this on faith as it was probably done earlier in the book. Then $\mathrm{Tor}_{p+1}^{U\mathfrak{g}}(R, R)$ is the $(p+1)$th homology group of the new complex

$$\cdots \longrightarrow (U\mathfrak{g} \otimes_R \Lambda^2 \mathfrak{g}) \otimes_{U\mathfrak{g}} R \xrightarrow{d_2 \otimes 1} (U\mathfrak{g} \otimes_R \Lambda^1 \mathfrak{g}) \otimes_{U\mathfrak{g}} R \xrightarrow{d_1 \otimes 1} (U\mathfrak{g} \otimes_R R) \otimes_{U\mathfrak{g}} R \longrightarrow 0.$$

Of course the zeroth degree term is just R, and the first degree term is simply

$$(U\mathfrak{g} \otimes_R \Lambda^1 \mathfrak{g}) \otimes_{U\mathfrak{g}} R \cong (U\mathfrak{g} \otimes_R R^n) \otimes_{U\mathfrak{g}} R \cong U\mathfrak{g}^n \otimes_{U\mathfrak{g}} R \cong R^n \cong \mathfrak{g} \cong \Lambda^1 \mathfrak{g}.$$

Since the exterior power of a free module is free, this process generalizes immediately yielding $(U\mathfrak{g} \otimes_R \Lambda^p \mathfrak{g}) \otimes_{U\mathfrak{g}} R \cong \Lambda^p \mathfrak{g}$. Hence our complex is simply $\Lambda^\bullet \mathfrak{g}$. We now need to compute the differential in this complex. Explicitly, the isomorphism $U\mathfrak{g} \otimes_R \Lambda^p \mathfrak{g} \otimes_{U\mathfrak{g}} R$ is given by $u \otimes \omega \otimes r \mapsto (u \cdot r)\omega$. Thus we define the isomorphic chain complex

$$\cdots \longrightarrow \Lambda^2 \mathfrak{g} \xrightarrow{\widetilde{d}_2} \Lambda^1 \mathfrak{g} \xrightarrow{\widetilde{d}_1} \Lambda^0 \mathfrak{g} \longrightarrow 0$$

where

$$\widetilde{d}_p(x_1 \wedge \cdots \wedge x_p) = \sum_{i=1}^{p} (x_i \cdot 1) x_1 \wedge \cdots \wedge \widehat{x_i} \wedge \cdots \wedge x_p.$$

We now need to compute the homology of this chain complex to obtain the Lie algebra homology of $\mathfrak{g}$. Since R is the trivial $\mathfrak{g}$-module, we have that $x_i \cdot 1 = 0$, so $\widetilde{d}_p(x_1 \wedge \cdots \wedge x_p) = 0$, meaning the differentials are all trivial, so the chain complex is isomorphic to its homology. Thus $H_p^{\mathrm{Lie}}(\mathfrak{g}, R) \cong \Lambda^p \mathfrak{g}$. ■

We now consider cohomology. Using the same idea as before, we start with the short exact sequence

$$0 \longrightarrow \mathfrak{J} \longrightarrow U\mathfrak{g} \longrightarrow R \longrightarrow 0.$$

Applying $\mathrm{Ext}^\bullet_{U\mathfrak{g}}(-, M)$ to this, and using contravariance of Hom in the first variable, yields the long exact sequence

$$\cdots \longrightarrow \mathrm{Ext}^{n-1}_{U\mathfrak{g}}(\mathfrak{J}, M) \longrightarrow \mathrm{Ext}^n_{U\mathfrak{g}}(R, M) \longrightarrow \mathrm{Ext}^n_{U\mathfrak{g}}(U\mathfrak{g}, M) \longrightarrow \mathrm{Ext}^n_{U\mathfrak{g}}(\mathfrak{J}, M) \longrightarrow \cdots.$$

Since $\mathrm{Hom}_{U\mathfrak{g}}(U\mathfrak{g}, -) \cong 1$ by identifying $U\mathfrak{g}$-linear maps $\varphi : U\mathfrak{g} \to M$ with $\varphi(1)$, the $\mathrm{Ext}^\bullet_{U\mathfrak{g}}(U\mathfrak{g}, M)$ modules vanish, yielding isomorphisms

$$\mathrm{Ext}^{n-1}_{U\mathfrak{g}}(\mathfrak{J}, M) \cong \mathrm{Ext}^n_{U\mathfrak{g}}(R, M) \cong H^n_{\mathrm{Lie}}(\mathfrak{g}, M)$$

for $n \geq 2$, and in degree 1 yielding the exact sequence

$$0 \longrightarrow \mathrm{Hom}_{U\mathfrak{g}}(R, M) \longrightarrow \mathrm{Hom}_{U\mathfrak{g}}(U\mathfrak{g}, M) \longrightarrow \mathrm{Hom}_{\mathfrak{U}\mathfrak{g}}(\mathfrak{J}, M) \longrightarrow \mathrm{Ext}^1_{U\mathfrak{g}}(R, M) \longrightarrow 0,$$

where we of course use $\mathrm{Ext}^0_{U\mathfrak{g}} \cong \mathrm{Hom}_{U\mathfrak{g}}$. Note that $\mathrm{Hom}_{U\mathfrak{g}} \cong \mathrm{Hom}_{\mathfrak{g}}$, and thus

$$\mathrm{Hom}_{U\mathfrak{g}}(R, M) \cong \mathrm{Hom}_{\mathfrak{g}}(R, M) \cong M^{\mathfrak{g}},$$

where the final isomorphism was seen earlier. We also already saw $\mathrm{Hom}_{U\mathfrak{g}}(U\mathfrak{g}, -) \cong 1$, so $\mathrm{Hom}_{U\mathfrak{g}}(U\mathfrak{g}, M) \cong M$. Finally, $\mathrm{Ext}^1_{U\mathfrak{g}}(R, M) \cong H^1_{\mathrm{Lie}}(\mathfrak{g}, M)$ yields the exact sequence

$$0 \longrightarrow M^{\mathfrak{g}} \longrightarrow M \longrightarrow \mathrm{Hom}_{\mathfrak{g}}(\mathfrak{J}, M) \longrightarrow H^1_{\mathrm{Lie}}(\mathfrak{g}, M) \longrightarrow 0.$$

To describe H^1, we study $\mathrm{Hom}_{\mathfrak{g}}(\mathfrak{J}, M)$.

Definition 1.4.8. Let M be a $\mathfrak{g}$-module. A *derivation* from $\mathfrak{g}$ into M is a R-linear map $D : \mathfrak{g} \to M$ such that the Leibniz rule holds

$$D[x, y] = x(Dy) - y(Dx).$$

The set of all derivations is denoted $\mathrm{Der}(\mathfrak{g}, M)$; it is an R-submodule of $\mathrm{Hom}_R(\mathfrak{g}, M)$. Note that if M is a trivial $\mathfrak{g}$-module, then $\mathrm{Der}(\mathfrak{g}, M) \cong \mathrm{Hom}_R(\mathfrak{g}^{ab}, M)$. This follows since if M is trivial, then $D[x, y] = xDy - yDx = 0$, so D factors through $\mathfrak{g}^{ab}$. The converse direction of the correspondence is immediate.

Example 1.4.9. If $m \in M$, deefine $D_m(x) = xm$. This is a derivation:

$$D_m[x, y] = [x, y]m = x(ym) - y(xm) = xD_my - yD_mx.$$

These derivations are called the *inner derivations* of $\mathfrak{g}$ into M, and they form a R-submodule $\mathrm{Der}_{\mathrm{Inn}}(\mathfrak{g}, M)$ of $\mathrm{Der}(\mathfrak{g}, M)$.

Example 1.4.10. Let $\varphi : \mathfrak{J} \to M$ be a $\mathfrak{g}$-linear map. Then the map $D_\varphi : \mathfrak{g} \to M$ defined by $D_\varphi(x) = \varphi(i(x))$ is also a derivation; here of course $i : \mathfrak{g} \hookrightarrow U\mathfrak{g}$ factors through $\mathfrak{J}$. We write $i(x) = x$ without loss of generality. This too is a derivation: since the bracket in $U\mathfrak{g}$ is the commutator and φ is $\mathfrak{g}$-linear,

$$D_\varphi[x, y] = \varphi[x, y] = \varphi(xy - yx) = x\varphi(y) - y\varphi(x) = xD_\varphi y - yD_\varphi x.$$

We will be able to show in this section that *all* derivations are of this form.

Lemma 1.4.11. *With notation as in example 1.4.10, the map $\varphi \mapsto D_\varphi$ is a natural isomorphism of R-modules:*

$$\operatorname{Hom}_{\mathfrak{g}}(\mathfrak{J}, M) \cong \operatorname{Der}(\mathfrak{g}, M).$$

Proof. Naturality is immediate. For the proof of isomorphism, we use the fact that the product map $U\mathfrak{g} \otimes_R \mathfrak{g} \to (U\mathfrak{g})\mathfrak{g}$ is epi, as was established in the proof of corollary 1.3.3. Write this map as g. Since $[x, y]$ becomes $xy - yx$ in $U\mathfrak{g}$, we have that

$$g(u \otimes [x, y]) = u[x, y] = uxy - uyx.$$

Thus we have

$$\operatorname{Ker} g \cong \langle u \otimes [x, y] - ux \otimes y + uy \otimes x \mid u \in U\mathfrak{g}; x, y \in \mathfrak{g} \rangle .$$

We write $ux \otimes y$ instead of $u \otimes xy$ since xy is not defined in $\mathfrak{g}$, but under f we have $f(ux \otimes y) = uxy$, so it is immaterial.

Let D be a derivation. Define the map $f : U\mathfrak{g} \otimes_R \mathfrak{g} \to M$ by $f(u \otimes x) = u(Dx)$. Since D is a derivation,

$$f(u \otimes [x, y] - ux \otimes y + uy \otimes x) = uD[x, y] - uxDy + uyDx = 0.$$

Thus f factors through $U\mathfrak{g}/\operatorname{Ker} g \cong \mathfrak{J}$, inducing a map $\varphi : \mathfrak{J} \to M$. We claim this is a map of $\mathfrak{g}$-modules. ■

Bibliography

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